

Birla Institute of Technology and Science, Pilani

Work Integrated Learning Programmes Division

M.Tech. in Artificial Intelligence and Machine Learning.

I Semester 2025-26

Course Number	AIMLZC416	
Course Name	Mathematical Foundations for Machine Learning	
Nature of Exam	Closed Book	# Pages 3
Weightage for grading	30%	# Questions 7
Duration	120 minutes	
Date of Exam	//2025 (FN: 09:00 - 11:00)	

Instructions

1. All questions are compulsory
 2. Marks are indicated against each question
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Q1. One definition of matrix multiplication is that when two matrices A and B are multiplied to get C , the i th row of C is obtained by a linear combination of the rows of B with the combining coefficients coming from the i th row of A . Let A and B be two lower-triangular $n \times n$ matrices such that the diagonal of B contains only non-zero values. If we know that $C = AB$ is a lower-triangular matrix, can you use the definition given here about matrix multiplication, to show that A must also be a lower triangular matrix? (4 marks)

Q2. We are given a matrix $m \times n, m > 1000, n > 1000$ matrix A consisting of $mn - 2$ zeroes and 2 non-zero values. Let the first non-zero entry be a_{pq} and the second one be b_{rs} . What is the maximum possible rank of the matrix A and what is the minimum value of $|p - r| + |q - s|$ when the matrix A attains its maximum rank? Give appropriate reasoning in your answer. (4 marks)

Q3. Consider the following matrix C

$$C = \begin{bmatrix} 6 & -33 & 19 \\ 6 & -8 & \frac{4}{3} \\ 9 & -33 & 16 \end{bmatrix}$$

Without direct multiplication of matrices

- i) Calculate the Trace of the matrix C_1 where $C_1 = C^7$ (1.5 marks)
- ii) Calculate the determinant of the matrix C_2 where $C_2 = C^8$ (1.5 marks)

Q4.(a) Assume that you are given a matrix $A \in \mathbb{R}^{n \times n}$ where $A = [a_{ij}]$ ($1 \leq i, j \leq n$). The matrix A has the following interesting property :

$$a_{i1} + a_{i2} + \dots + a_{in} = 0 \quad \text{for each of } i = 1, 2, \dots, n.$$

- i) Can zero be one of the eigenvalues of this matrix ? Give reasoning for your answer. (1 mark)
- ii) Derive the determinant of this matrix. (1 mark)
- iii) Derive at least one eigenvector-eigenvalue pair of this matrix (1 mark) .

(b) Consider a vector $\mathbf{r} \in \mathbb{R}^k$. Here $k > 132$. Consider a matrix $\mathbf{M}$ defined as follows : $\mathbf{M} = \mathbf{r}\mathbf{r}^T$. Based on this definition answer the following :

- i) How many non-zero eigenvalue does the matrix $\mathbf{M}$ have ?. Derive all the nonzero eigenvalues. (1 mark)
- ii) Find the sum of absolute value of eigenvalues of this matrix i.e $\sum_{i=1}^k |\lambda_i|$ where λ_i is the i^{th} eigenvalue of $\mathbf{M}$. (1 mark)

Q5. In training a machine learning model, the loss function is defined by

$$L(w_1, w_2) = \ln(1 + w_1^2 + w_2^2),$$

where w_1 and w_2 are the model parameters.

- a) Derive the second-order Taylor expansion of $L(w_1, w_2)$ around the current parameter estimate $(0, 0)$. (2 marks)
- b) Using the Taylor polynomial, estimate the loss value at the point $(w_1, w_2) = (0.1, -0.2)$. (1 mark)
- c) Compute the exact value of $L(0.1, -0.2)$ and calculate the absolute error of the approximation. (1 mark)

Q6. i) Discuss the benefits of applying the matrix reshape concept when differentiating a matrix with respect to a vector, especially from a programming implementation perspective, and provide a justification for its use. (1 mark)

ii) Find the derivative of the matrix $K := BB^T$ with respect to the matrix B , where $B \in \mathbb{R}^{2 \times 3}$ is given by:

$$B = \begin{bmatrix} -x_1 & x_2^2 & x_1x_2 \\ x_1^1 + x_2^2 & 2.5x_2 & 0.7x_1 \end{bmatrix}$$

(3 marks)

Q7 A real $n \times n$ matrix G is called positive if it is symmetric and $x^T G x \geq 0$ for all $x \in \mathbb{R}^n$.

- a) Comment on the connection between positive definite matrices and positive matrices. (1 mark)
- b) Let G denote an $n \times n$ positive matrix
 - (i) If A is any $m \times n$ matrix and G is positive, then prove or disprove $A^T G A$ is positive. (1 mark)
 - (ii) If $G = \text{diag}(d_1, d_2, \dots, d_n)$ and each $d_i \geq 0$, then prove or disprove G is positive. (1 mark)
- c) If $A = U \Sigma V^T$ is the Singular value decomposition (SVD) of A , what kind of a matrix is $W = UV^T$? (1 marks)
- d) If A is an $n \times n$ matrix, a factorization $A = GQ$ where G is positive and Q is orthogonal is called a polar decomposition of A . Prove or disprove using SVD that every square real matrix has a polar form. (2 marks)